CS4423 - NETWORKS: SUMMARY

1. Concepts, Definitions

- Graph (simple, directed, multigraph)
- Adjacency matrix
- Bipartite graph
- Node degree, Neighbors of a node, social graph
- Relations: reflexive, symmetric, transitive
- Paths, simple paths, cycles, path length
- Connected components
- Trees, forests
- Random tree
- Graph Automorphisms
- Distance, shortest path
- Spanning tree
- Degree centrality
- Eigenvector centrality
- Closeness centrality
- Betweenness centrality
- Weakly connected components vs Strongly connected components
- Binomial distribution, Poisson distribution
- Degree distribution
- Giant component
- Characteristic path length
- Node clustering coefficient, graph clustering coefficient
- Graph transitivity
- Small world behavior
- (n, d)-circle graph
- Power law distribution

2. Facts, Theorems

- Handshake Lemma
- Composition of relations; their adjacency matrix
- Characterisation of bipartite graphs
- Projections of bipartite graphs; their adjacency matrix
- Cayley's Formula
- Perron-Frobenius Theorem
- Properties of binomial and Poisson distribution
- Erdös-Rényi Theorem on appearance of giant components
- Partial order of strongly connected components
- Bow-Tie structure of the WWW
- Spectral analysis of hubs and authorities
- Molloy–Reed criterion
- Degree distribution of a BA model

3. Procedures, Algorithms

- Prüfer Code; Random Trees
- DFS
- BFS and its variants
- Orbit Algorithm
- Shortest paths
- Computing betweenness centrality
- Knuth's algorithm S
- ER random graph generators $G(n, m)$ and $G(n, p)$
- Hubs and Authorities
- Page Rank
- Watts-Strogatz
- Discrete Power Law Distribution
- Configuration Model
- BA model of linear preferential attachment

4. Not on the Exam ... but useful:

- Programming `python`: lists, dictionaries, comprehension, functions, arithmetic
- Using the commands from the `networkx` package
- Computing with `numpy` arrays
- Plotting data with `pandas` and `matplotlib`
- Working with `jupyter` Notebooks